

EE1101: Circuits and Network Analysis - Midterm Exam

Name: _____

Roll No: _____

Instructions

1. The exam lasts for 90 minutes, beginning at 14:30. Please ensure you submit your paper by 16:00. Note that late submissions may incur penalties.
2. The exam consists of two sections. Section I includes straightforward questions, and you should write your answers in the space next to each question. Answers written elsewhere will not be considered for grading and will be evaluated in a binary manner. For Section II, please write your solutions in the answer book.
3. Carefully review all questions. Misinterpretations will not be considered. Make sure to write your answers in the designated spaces for each question; answers submitted outside these areas will not be accepted.
4. Adopting unfair practices will lead to an F grade and will be reported to the disciplinary committee.

I hereby declare that I have read all the instructions carefully and will adhere to them.

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Section I (5 Points)

1. (2 points) Analyze the BJT circuit presented in Fig. 1(a) by using the equivalent circuit in Fig. 1(b). Calculate the current i_e . Please provide your answer here: _____.

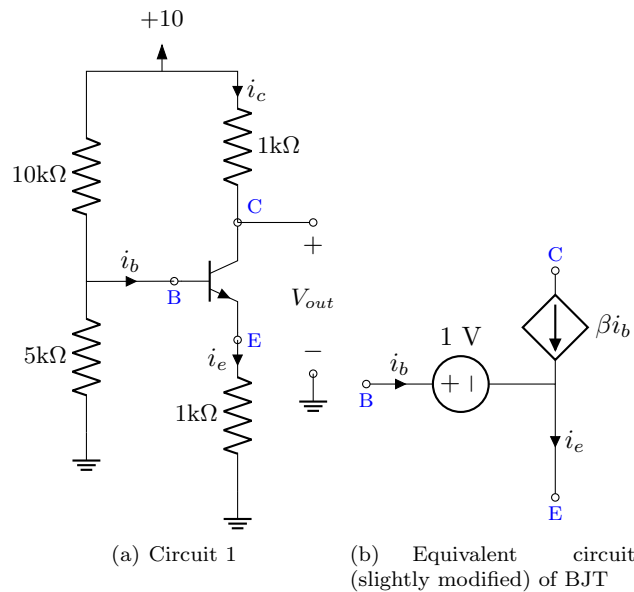


Fig. 1: Circuits with BJT (Questions 1 and 2)

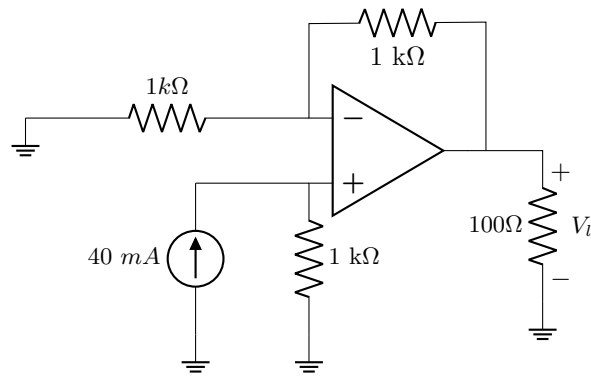


Fig. 2: Circuit for question 2

2. (1 point) Calculate the output voltage V_l for the circuit illustrated in Fig. 2. Assume the op-amp is ideal. Please provide your answer here: _____.
3. (1 point) Given the phasor representations of two signals, $v_a(t)$ and $v_b(t)$, with a frequency of 50 Hz, where $\mathbf{V}_a = 10\angle\frac{\pi}{2}$ and $\mathbf{V}_b = 2\angle\frac{-\pi}{4}$, calculate the time difference between the positive peaks of these two signals. Please provide your answer here: _____.
4. (1 point) For the circuit shown in Fig. 3, it is desired to add a circuit element in parallel to the inductor such that the reactive power associated with the source is 0. Comment on the nature of the element and its value. Please provide your answer here: _____.

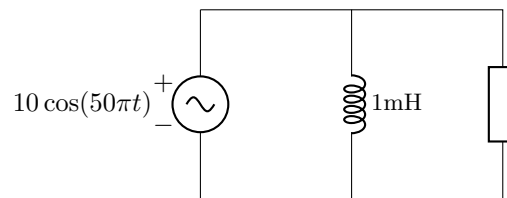


Fig. 3: Circuit for question 4

Section II (15 Points)

1. (5 points) For the circuit shown in Fig. 4, derive an expression for the inductor current i_l . Assume that t_0 is large enough for the circuit to reach steady state prior to t_0 and the current through the inductor at $t = 0$ is 0.

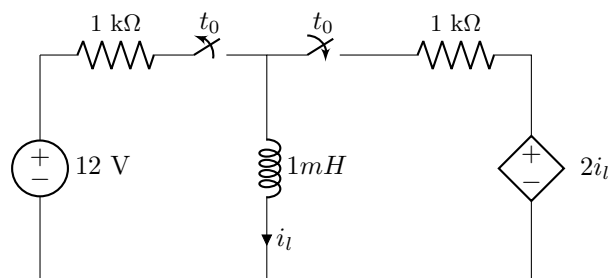


Fig. 4: Circuit for question 1

2. (5 points) The average and RMS values of the signal $v(t)$ (shown in Fig. 5) are 20 and $\frac{40}{\sqrt{3}}$ respectively. Compute the RMS of the signal $v_2(t) = 1 + v(t)$.

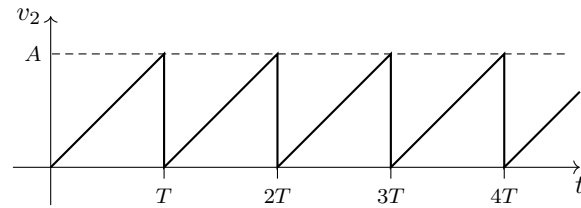


Fig. 5: Waveform for question 2

3. (5 points) For the circuit shown in Fig. 6, compute the open circuit voltage and short circuit current associated with the port $p - o$.

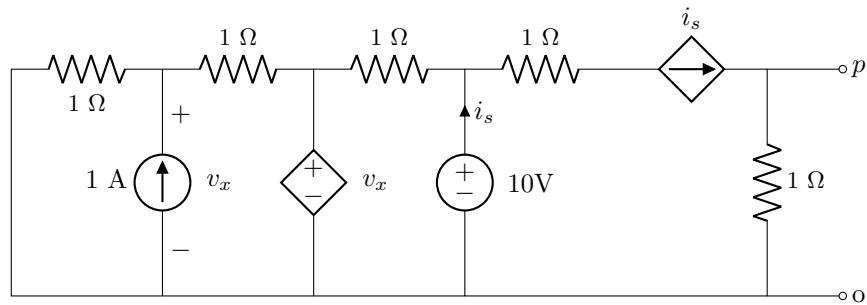


Fig. 6: Waveform for question 3

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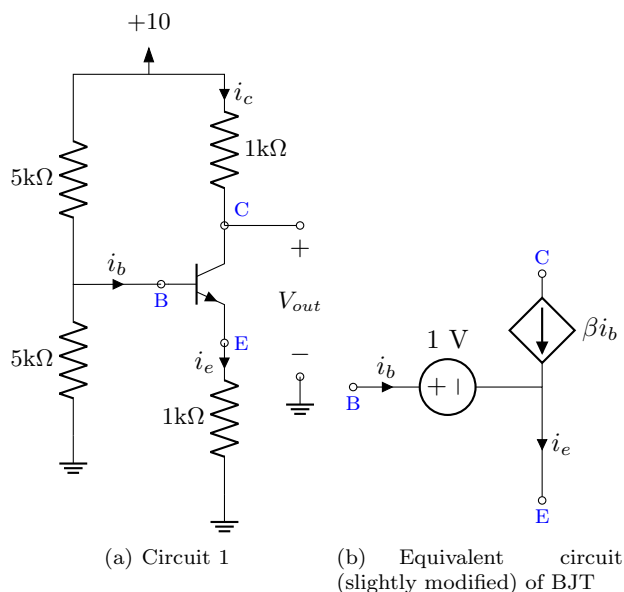


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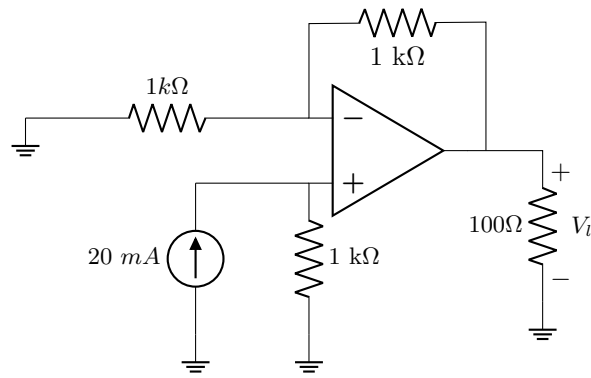


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2. (1 point) Calculate the output voltage V_l for the circuit illustrated in Fig. 2. Assume the op-amp is ideal. Please provide your answer here: _____.
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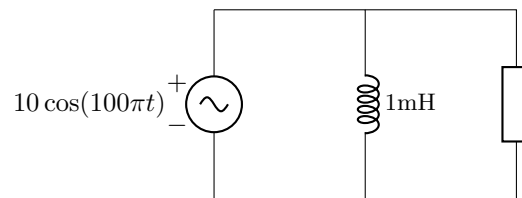


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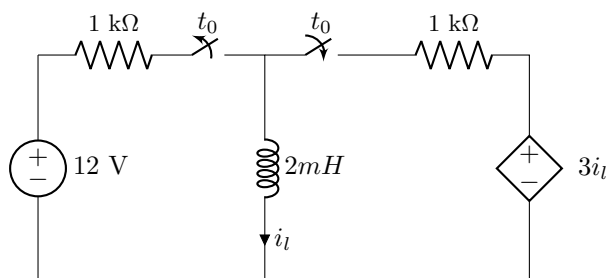


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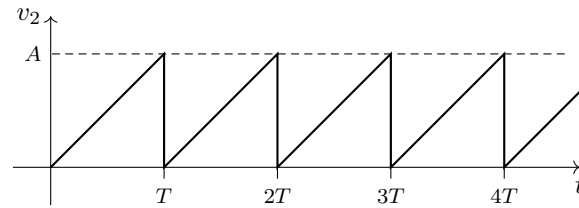


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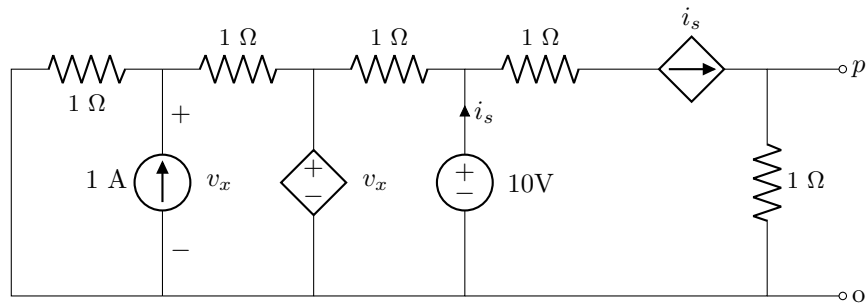


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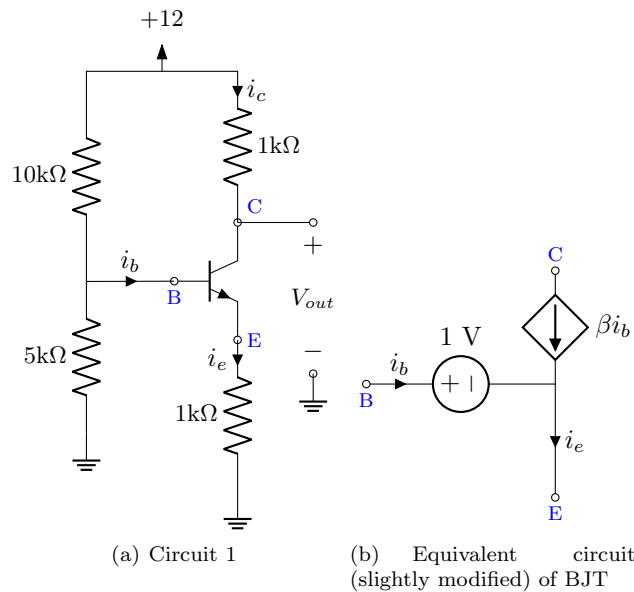


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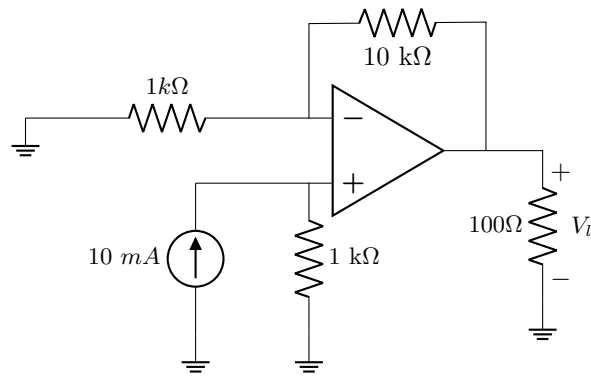


Fig. 2: Circuit for question 2

2. (1 point) Calculate the output voltage V_l for the circuit illustrated in Fig. 2. Assume the op-amp is ideal. Please provide your answer here: _____.
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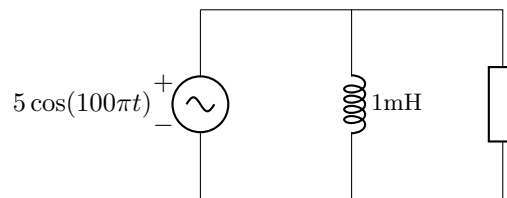


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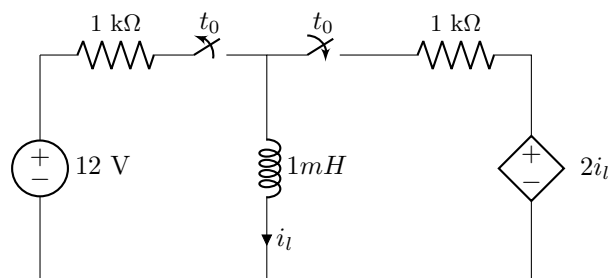


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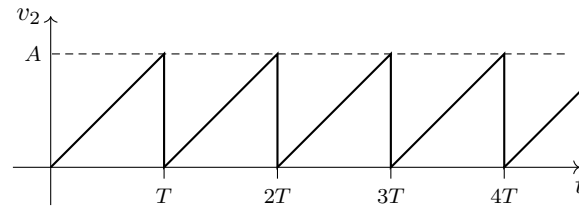


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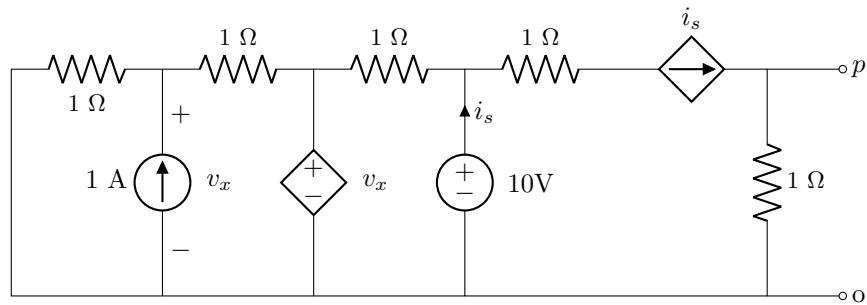


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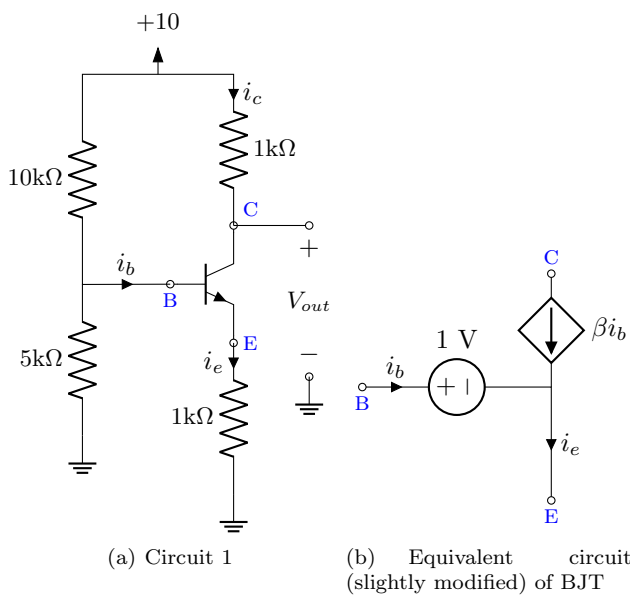


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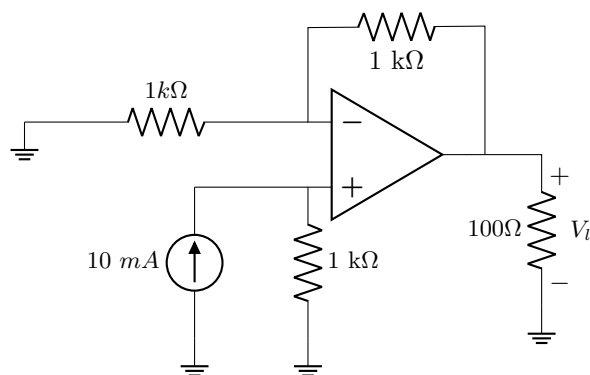


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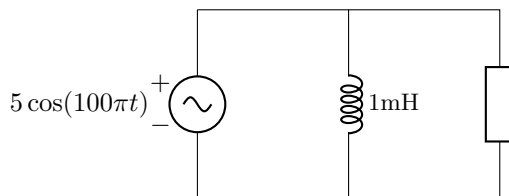


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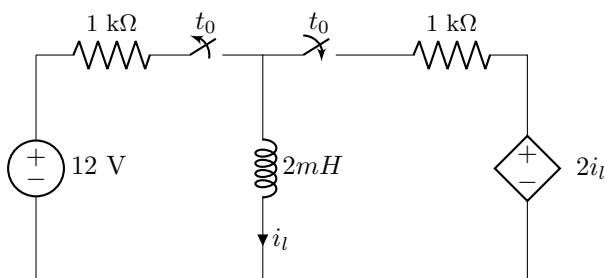


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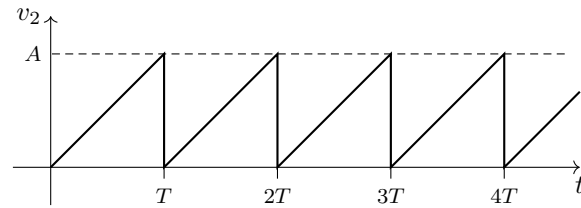


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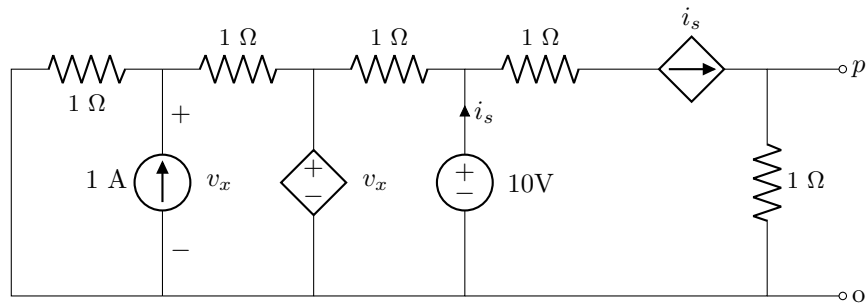


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